

Exploring Medical Insurance Charges

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1. Introduction

For this project, I used the Medical Insurance dataset, which contains demographic and health-related information for individuals, including age, sex, BMI, number of children, smoking status, region, and the final medical charges billed to the insurance provider. Each record represents one person and their associated health and lifestyle characteristics.

My goal was to answer three main questions:

- (1) What factors make an individual's medical charges high?
- (2) Which features are most strongly connected to insurance cost?
- (3) Are there any surprising patterns or relationships hidden in the data?

2. Cleaning Summary

The raw insurance dataset required several preprocessing steps to ensure data quality before analysis:

- **Missing values:** An initial inspection showed that all seven columns—age, sex, BMI, children, smoker, region, and charges—contained no missing values. Because the dataset was fully complete, no imputation or row removal was necessary.
- **Data types:** The BMI and charges columns were explicitly converted to numeric using `pd.to_numeric()` to guard against any hidden formatting issues. All other columns were already stored in appropriate types: integers for age and children, floats for BMI and charges, and object types for categorical variables such as sex, smoker, and region.
- **Duplicates:** A check for duplicate rows revealed a small number of repeated entries. These were removed using `df.drop_duplicates()`, ensuring that each record represented a unique individual.
- **Outliers:** The *charges* column exhibited a heavy right tail, with several extreme values exceeding \$48,000. A boxplot confirmed the presence of significant outliers. To reduce distortion in later modeling steps, charges were capped at the 99th percentile using the `clip()` method. This preserved the overall distribution while preventing extreme values from dominating the analysis. **These outliers were not removed because** they represent real individuals with genuinely high medical expenses, especially smokers and older adults, and excluding them would remove meaningful variation from the dataset.

- **Validation check:**

To ensure the dataset was fully clean, three assertions were applied:

- all values in the dataset were non-null,
- all charges were strictly positive,
- the dataset contained exactly seven columns.

All checks passed successfully.

All cleaning steps were wrapped into a reusable `clean_data()` function to maintain consistency and reproducibility throughout the project.

3. Feature Engineering Summary

- **One-hot encoding for categorical variables:** The dataset contains three categorical features—sex, smoker, and region. Since these variables have no natural ordering, I applied one-hot encoding to convert them into numerical indicators. This allows machine-learning models to interpret category membership without imposing artificial order

- **Scaling numerical features:** The numerical columns *age* and *BMI* were standardized using `StandardScaler`. Scaling ensures that features measured on different ranges contribute proportionally during model training and prevents large-scale variables from dominating the learning process.

- **bmi_per_age (BMI divided by age):** This ratio feature captures how an individual's BMI compares relative to their age. It helps highlight unusually high BMI values among younger individuals, which may be associated with elevated medical risk.

- **bmi_age_interaction (BMI × age):** This interaction feature models the idea that the combined effect of age and BMI may influence medical charges more strongly than either variable alone. Older individuals with high BMI often incur higher healthcare costs, and this feature helps represent that relationship.

- **Log-transformed charges:** The raw *charges* distribution was heavily right-skewed. Applying `np.log1p()` produced a more symmetric distribution, which is beneficial for modeling and reduces the influence of extreme values.

- **Age bins (Young, Middle, Senior):**

To capture non-linear age effects, I grouped individuals into three age categories using fixed-width bins:

- Young (17–29),

- Middle (30–49),
- Senior (50+).

This transformation simplifies comparisons across age groups and helps reveal broader demographic patterns.

• **Redundant feature check (Correlation > 0.95):** To detect redundant features, I computed the feature–feature correlation matrix. No real features exceeded the 0.95 threshold, indicating that the dataset does not contain duplicate or overly similar information. A temporary synthetic feature (*bmi_copy*) was created only to demonstrate how highly correlated features appear; it was removed afterward.

After creating all engineered features, the dataset was clean, enhanced, and ready for modeling.

4. Key Findings

Finding 1 — Smoking status is the strongest predictor of medical charges.

The boxplot comparing smokers and non-smokers showed the most dramatic separation in the dataset. Smokers had a much higher median charge and a significantly wider spread, indicating large variability in medical expenses. The correlation matrix confirmed this: *smoker_yes* had the strongest positive correlation with both raw and log-transformed charges. This aligns with medical expectations — smoking is a major risk factor that substantially increases healthcare costs.

Finding 2 — BMI influences charges, especially at higher values.

The BMI histogram showed a right-skewed distribution, with most values between 25 and 35. The scatter plot of BMI vs. log-transformed charges revealed a clear upward trend, particularly for BMI values above 30 (the obesity threshold). Smokers with high BMI formed a distinct cluster at the highest charge levels, showing that BMI interacts with lifestyle factors to amplify medical costs.

Finding 3 — Age has a strong, non-linear effect on medical expenses.

The age histogram showed a unimodal distribution with a peak at age 18, but the groupby analysis revealed a clear upward trend in medical charges as age increases. Young adults had the lowest average charges, middle-aged individuals showed noticeably higher costs, and seniors had the highest medical expenses overall. This pattern reflects the expected rise in healthcare utilization with age and confirms that age is an important demographic factor influencing insurance charges.

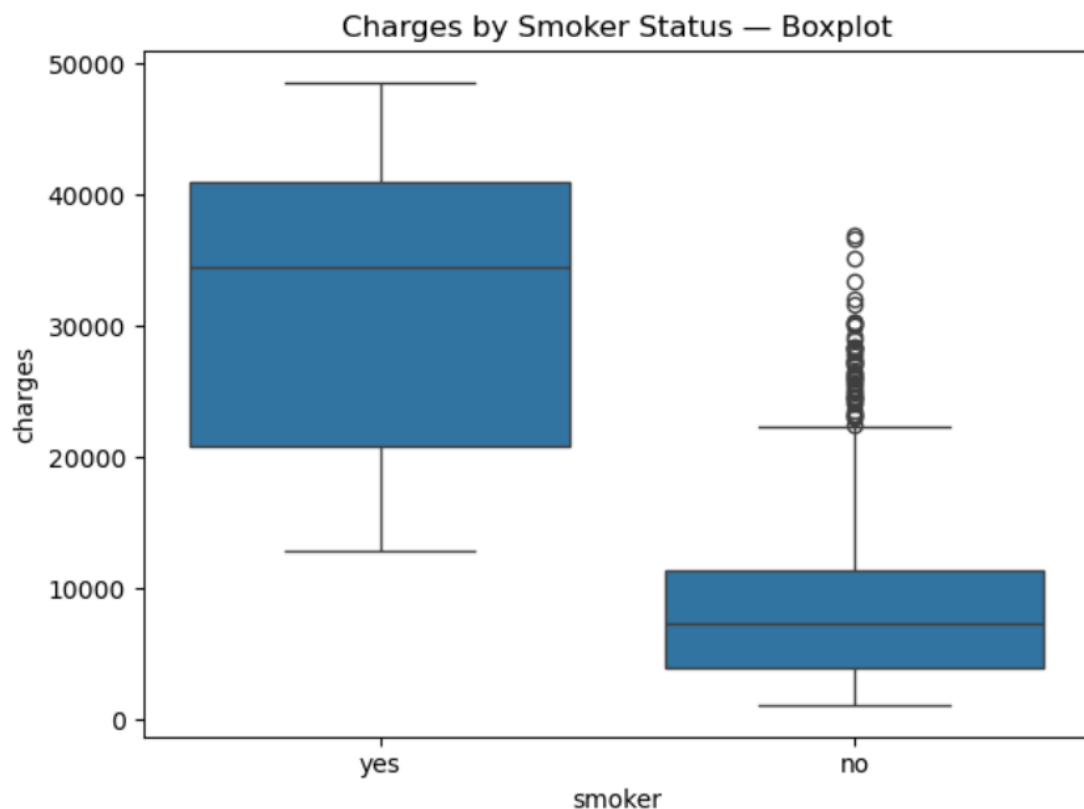
Finding 4 — Sex has little to no effect on medical charges.

The boxplot comparing males and females showed nearly identical distributions. Both groups had similar medians, interquartile ranges, and whiskers. Although females had a

slightly higher median, the difference was small and not meaningful. This indicates that sex is not a major predictor of medical charges compared to stronger factors such as smoking and BMI.

[Figure 1: Boxplot of Medical Charges by Smoking Status]

I selected this figure because it clearly shows the strongest relationship in the dataset. The difference between smokers and non-smokers is large and easy to see, making this plot the best visual evidence that smoking is a major driver of higher medical charges.



5. What I Would Do Next

If I had more time, I would:

- (1) train a simple linear regression model using my engineered features and evaluate how well it predicts medical charges;
- (2) explore potential interaction effects, such as whether the impact of BMI on charges differs between smokers and non-smokers; and
- (3) analyze how medical charges vary across regions to determine whether geographic factors introduce systematic differences that the model should account for.

End Of Report

